

Lab 7: Energy — Photosynthesis & Cellular Respiration

BIOL-1

Name: _____ Date: _____

Overview

This paper-based lab explores the flow of energy through living systems. You will analyze the complementary processes of photosynthesis and cellular respiration, trace the path of carbon and energy, and solve problems related to ATP production.

Learning Objectives

By the end of this lab, you will be able to:

1. Compare and contrast the chemical equations for photosynthesis and cellular respiration.
2. Trace the movement of electrons through the electron transport chains in chloroplasts and mitochondria.
3. Calculate the theoretical ATP yield from glucose under aerobic and anaerobic conditions.
4. Explain the relationship between autotrophs (producers) and heterotrophs (consumers).

Materials

- Worksheet: Energy Pathways
- Colored pencils/markers
- Cut-out models of molecules (glucose, CO₂, O₂, H₂O, ATP)

Activity 1: The Carbon Cycle Simulation

Using the cut-out models, simulate the cycling of carbon atoms:

1. **Photosynthesis:** Build a glucose molecule from CO₂ and water (using "light energy").
2. **Respiration:** Break down the glucose molecule back into CO₂ and water, releasing "energy" (ATP).

3. **Track:** Where did the Carbon go? Where did the Oxygen go? Where did the Hydrogen go?

Activity 2: Electron Transport Chain Diagramming

Draw detailed diagrams of the ETC in both the chloroplast (thylakoid membrane) and mitochondrion (inner membrane).

- Label protons (H^+), electrons (e^-), ATP synthase, and the electron carriers (NADH, NADPH, $FADH_2$).
- Show the direction of H^+ pumping and ATP synthesis.

Activity 3: Fermentation Case Studies

Analyze scenarios where oxygen is limited (e.g., sprinting muscle cells, yeast in a sealed bottle).

- Determine which pathway is used (Lactic Acid or Alcoholic Fermentation).
- Calculate the energy yield compared to aerobic respiration.

Conclusion

Summarize how energy flows (one way: Sun $\rightarrow$ Earth $\rightarrow$ Heat) while matter cycles (Carbon/Oxygen/Hydrogen loop between photosynthesis and respiration).